

Staking Your Crypto: What are the Stakes?

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The use of crypto-assets as an investment asset to generate passive income has become steadily more popular since the implementation of smart contracts on blockchains becoming popular with Ethereum in 2017. It reached an interim all-time high with the massive price fluctuations of crypto-assets in 2021. Many of these ways of generating returns on crypto-assets are an integral part of Decentralized Finance (DeFi) applications. But there are also risks involved. Significant macroeconomic headwinds and the cooling down of capital markets has recently put a persistent strain on the liquidity of crypto businesses. It is therefore not surprising that some of them are struggling economically, while others have already gone bankrupt. The winding up or liquidation of a crypto business has a significant impact on the value of assets and carries the great risk that crypto savers are left with pennies on the dollar. If crypto-assets are really valuable, one might think about visiting a lawyer and ultimately go to court. But what are the rights the former crypto-asset holder has in the event of its investment intermediary's insolvency? This is the question this article seeks to address.

A. The Premise

Cryptos are not capital assets and do not earn interest – or so one thought. Although highly unlikely that ‘Satoshi Nakamoto’ ever contemplated use of Bitcoin as a traditional investment asset, the practice has steadily been increasing in popularity. A wide range of terms are floated about, but the basic underlying idea is the same: you give up at least partial control over your crypto-assets² in favour of some other party who uses it in the crypto market to generate profits, of which you receive a share. In common crypto parlance, you ‘stake’ your cryptoassets to the other party’s particular investment strategy, and you receive an ‘annual percentage yield.’ Such returns

¹ All University of Vienna. We thank Amy Held for her contribution to this article.

² In the following, the term crypto-asset shall be used as a reference for any type of financial value transferred by means of a blockchain. The concept of crypto-asset generally purports to refer to a wide array of diverse coins or tokens that very often blur the distinctions between traditional financial assets’ classes. It is however worth keeping in mind that, while the broader concept of crypto-asset is used in this paper, it is coins (e.g. Bitcoin or Ether) and stablecoins (e.g. U.S.-Dollar Coin (U.S.DC)) that are mostly used today for generating returns via staking, yield farming, or liquidity mining.

sometimes reach hundreds or thousands of percentage points and can, accordingly, be very attractive.³

But equally high are the risks involved. Some are inherent in the blockchain protocol itself, often known as 'on-chain' risks, such as flaws in the underlying code. Others arise from malicious behaviour on the part of the developers or other crypto users, which are also known as 'off-chain' risks.⁴ Of these, 'rug pulls' are perhaps the most harmful, as they undermine the overall trust of people in the crypto economy and the blockchain technology itself. Rug pulls are pure and simple scams that fraudulently siphon off all the funds that investors, lured by extremely attractive promises of financial returns, have contributed to phony platforms.

However, no risk lurks with such omnipresence as that classic risk against which no commercial enterprise can ever wholly rest assured (and which a great deal of commercial law is concerned to allocate *ante ex*): counterparty insolvency. With significant macroeconomic headwinds and the cooling down of capital markets, market participants within the crypto space have recently been facing a persistent strain on their business liquidity.⁵ Whilst such participants claim that all crypto-holdings are safe, their clients and account-holders feel rather differently and have begun withdrawing their assets. This only worsens the situation for already tarnished crypto

³ Most crypto trading platforms offer rather moderate interest rates, see Binance Earn, available at: <www.binance.com/en/earn> (last visited 21 December 2022); Coinbase Earn, available at: <www.coinbase.com/earn> (last visited 21 December 2022); Kraken Earn, available at: <www.kraken.com/features/staking-coins> (last visited 21 December 2022); Bitpanda Staking, available at: <www.bitpanda.com/de/staking> (last visited 21 December 2022); other platforms, on the other hand, lure with astronomical APY values. The annual APY of BCH-BNB of is approximately 1065871%, see CoinMarketCap, available at: <<https://coinmarketcap.com/de/yield-farming/>> (last visited 21 December 2022).

⁴ *Raphael Auer et al.*, 'The Technology of Decentralized Finance (DeFi)' (2023) BIS Working Papers No 1066, 3 (mentioning that the "rise of DeFi has been accompanied by many incidents with an accumulated total loss exceeding 3 billion USD."); *Liyi Zhou et al.*, 'Decentralized finance (defi) attacks' (2022) Cryptology ePrint Archive, Paper 2022/1773, available at: <<https://eprint.iacr.org/2022/1773>> (last visited 24 January 2023).

⁵ FSB, 'Regulation, Supervision and Oversight of Crypto-Assets Activities and Markets' (11 October 2022), 5; Board of Governors of the Federal Reserve System, Financial Stability Report (November 2022), 45-46; *Raphael Auer et al.*, 'The Technology of Decentralized Finance (DeFi)' (2023) BIS Working Papers No 1066, 2-3.

businesses. Hence some crypto companies have already gone insolvent;⁶ market trends indicate that others are likely to follow.⁷

We are in this paper not concerned with the legal nature of the crypto-asset itself, such as a Bitcoin or a DAI, which is itself very disputed. Rather, we assume that there is some protection for the holders of such assets are focussing on the problem whether it is foregone or lost by one of the operations under examination. Accordingly, the main purpose of this paper is to examine the rights a 'staker' (as our crypto investor will be henceforth be termed, notwithstanding the technical meaning of the term) might have in the event of the insolvency of their 'staking' intermediary. Are they entitled to make a direct claim to 'ownership' to 'their' staked assets – notwithstanding that they have conceded control over it, or even transferred it outright to the intermediary – such that they are removed from the intermediary's insolvent estate altogether? If not, are they to be treated as secured creditor, or at least, even a preferential creditor? Or will they simply have no other choice but to join the queue with the intermediary's other unsecured creditors for a *pari passu* share in whatever assets are left? Answering this

⁶ The year 2022 started off with the crash of the Terra Network, see *Joshua Oliver, Scott Chipolina and Eva Szalay*, 'Luna crash sends a chill through decentralised finance market' (2022), available at: <www.ft.com/content/c10bc6f7-abbe-45dc-9367-042186c3336f>, accessed 23.1.2023. In June 2022 the crypto hedge fund Three Arrows Capital was ordered to liquidate, see *Serena NG, Caitlin Ostroff and Vicky Ge Huang*, 'Crypto Hedge Fund Three Arrows Ordered by Court to Liquidate' (2022), available at: <www.wsj.com/articles/crypto-fund-three-arrows-ordered-to-liquidate-by-court-11656506404> (last visited 23 January 2023). In July the crypto broker Voyager Digital files for bankruptcy, see *PRNewswire*, 'Voyager Digital Commences Financial Restructuring Process to Maximize Value for All Stakeholders' (2022), available at: <www.prnewswire.com/news-releases/voyager-digital-commences-financial-restructuring-process-to-maximize-value-for-all-stakeholders-301581177.html> (last visited 23 January 2023); followed by the bankruptcy of the crypto lending platform BlockFi in November 2022, see *BlockFi*, 'Important Client Update', available at: <<https://blockfi.com/November28-ClientUpdate>> (last visited 23 January 2023). After pausing all withdrawals in June 2022, the crypto lending platform Celsius filed for bankruptcy in July 2022, see *Business Wire*, 'Celsius Network Initiates Financial Restructuring to Stabilize Business and Maximize Value for All Stakeholders' (2022), available at: <www.businesswire.com/news/home/20220713005911/en/Celsius-Network-Initiates-Financial-Restructuring-to-Stabilize-Business-and-Maximize-Value-for-All-Stakeholders> (last visited 23 January 2023); followed by the insolvency of the German cryptocurrency exchange Nuri, see *Paulina Okunyté*, 'Crypto Exchange Nuri Files for Insolvency in Germany, Celsius Bankruptcy to Blame', <<https://dailycoin.com/crypto-exchange-nuri-files-for-insolvency-in-germany-celsius-bankruptcy-to-blame/>> (last visited 11 August 2022). In November 2022 the crypto-exchange FTX files for bankruptcy, see *MacKenzie Sigalos*, 'Sam Bankman-Fried steps down as FTX CEO as his crypto exchange files for bankruptcy' (2022), available at: <www.cnbc.com/2022/11/11/sam-bankman-frieds-cryptocurrency-exchange-ftx-files-for-bankruptcy.html> (last visited 23 January 2023).

⁷ The market capitalisation of crypto-assets fell by more than 2/3 from September 2021 to December 2022 (2.892.035.226.113 USD on the 11 September 2021 to 797.051.559.613 USD on the 30 December 2022), see *CoinMarketCap*, 'Today's Cryptocurrency Prices by Market Cap', available at: <<https://coinmarketcap.com>> (last visited 23 January 2023).

seemingly simple question is no easy task; and in our view, this is for three main reasons.

First, a cursory view of the 'staking' market reveals a wide range of diverse practices, and it is erroneous to think that, simply because they have a basic economic objective in common (i.e., generate passive returns on a crypto-asset holding) that they will be treated alike as a matter of law. To the contrary, differences in practices lead to very different set of facts that courts will deem as material, and from which legal inferences and findings will be drawn when determining the rights of the staker in the intermediary's insolvency.

It is further not helpful that the promoters of staking schemes do not generally take legal advice before devising and advertising their investment strategies. Some adhere to the "Code is law"⁸ philosophy and consider that any problems arising from staking will be solved via the blockchain itself. Others, whilst recognising that their operations might be subject to legal processes, bandy about terms such as "property," "trust," and "ownership"⁹ without any consideration of what these mean – or require – in law. As every disputes lawyer will know, what the parties thought they were doing or intended to do, as evidenced by their written agreements and their conduct, on the one hand; and what they actually did in law, on the other, are often two very different things.

This is especially so in matters of proprietary rights – that holy grail of rights in any insolvency – which are not, as are contractual matters, a matter of the parties' intention and agreement. The rules of property law remain a mandatory form of private law from which parties cannot derogate; failure to comply with the requirements of the

⁸ See Primavera De Filippi and Aaron Wright, *Blockchain and the Law* (Harvard University Press 2018) 193-194, 199; originally, this idea originated in a book by Lawrence Lessing, however, he explicitly states that Code is Law does not mean that software and legislation are equivalent. Rather, this pithy formulation is meant to indicate that code, much like the other forces, regulates behaviour: ("Code is not law, any more than the design of an airplane is law"), see Lawrence Lessing, *Code: And Other Laws of the Cyberspace, Version 2.0* (Basic Books 2006), 324.

⁹ Gilead Cooper, 'Virtual property as trust assets and Feature investments' (2021) *Butterworths Journal of International Banking and Financial Law*, 751 (751-752); Helmut Stix, 'Ownership and purchase intention of crypto-assets – survey results' (2019), 29-31, available at: <www.ecb.europa.eu/pub/conferences/shared/pdf/20191126_payments_conference/academic_paper_stix.pdf> (last visited 23 January 2023); ZeMing M. Gao, 'Digital assets are subject to property laws' (2022), available at: <<https://coingeek.com/digital-assets-are-subject-to-property-laws/>> (last visited 23 January 2023); Thomson Reuters, 'Non-fungible Tokens: Asset ownership via blockchain rockets into legal' (2021), available at: <www.thomsonreuters.com/en-us/posts/legal/non-fungible-tokens-legal/> (last visited 23 January 2023); UK Law Commission, 'Digital Assets: Consultation Paper' (2022), available at: <www.lawcom.gov.uk/project/digital-assets/> (last visited 23 January 2023).

relevant property law will not result in the creation of a valid property right under that law, irrespective of what the parties purport to agree between themselves. Even so, contracts affect the property situation. For instance, when A agrees to "sell" his Bitcoin to B, this would normally entail a transfer of property (provided that the governing law recognizes Bitcoin as objects of property rights). In contrast, if A agreed to "lease" his Bitcoin to B, this could hardly be constructed as a transfer of property. Whether property passes will therefore depend on the correct characterization of the transaction. This characterization has to be done on the basis of the national law governing the contract. This law also provides the rules of interpretation a court will use to determine the parties' intentions.¹⁰

Second, the answer to what the parties did actually achieve in law is neither a straightforward issue. Viewed as a simple bilateral matter as between the parties, courts will look to a vast range of factors to determine their respective rights and obligations: primarily, their express and implied agreements, their conduct, and their compliance with any relevant legal formalities. From an international perspective, however, the matter becomes complicated because legal systems around the world accord different weight to these factors, and indeed, have different substantive requirements in respect of them. As a result, it may well be the case that the same set of facts leads to different conclusions as to the parties' respective rights, depending on which law is held to be the 'applicable law' to their relationship and transaction.

As a very simple example, Law A might hold that an agreement to create a security interest or any delivery of an asset pursuant to such agreement must be perfected by means of public registration; whereas Law B might require only a valid agreement plus delivery for the security interest to be validly constituted. Suppose, then, that a staking agreement purports to grant the staker a security interest in the intermediary's pool of staked assets, and obliges the intermediary to register the security 'in accordance with any applicable law' upon delivery by the staker of his assets to the intermediary to be added to the pool. However, upon such delivery by the staker, and in breach of their agreement, the intermediary fails to take any steps to register the security interest. Accordingly, the question of whether the security

¹⁰ See Lord Collins of Mapesbury and Jonathan Harris (eds), *Dicey, Morris & Collins The Conflict of Laws*, vol 2 (15th edn, Sweet & Maxwell 2012) para. 32-143 (2012); Ole Lando, III *Contracts Encyclopedia of Comparative Law* 24-212-24-213 (1971); this is sometimes also explicitly stated in the conflict-of-laws provisions, see e.g. Art 12(1)(a) Rome I Regulation.

interest was validly constituted in favour of the staker depends on whether Law A or Law B (or some other law) applies.

Identifying this applicable law in any given case is further complicated. It is a well-established rule of private international law that a bankruptcy court will generally apply the *lex fori*, i.e., its own domestic law, which is often known in the bankruptcy context as the *lex fori concursus*.¹¹ This principle is of paramount importance as it allows for the centralisation of the enforcement of all claims against the debtor – i.e., our now-insolvent staking intermediary – in a single tribunal. It is also a well-established rule that the *lex fori* applies to the characterisation of any issue in dispute for the purpose of the conflict of laws. In our above example, is the real issue between the parties a matter of breach of contract (and if so, what kind of contract), or one of property rights *in rem*? Courts in different jurisdictions will come to different conclusions on this issue, and in our case, characterisation will be determined by the bankruptcy court in accordance with its own domestic rules.

Third, once the bankruptcy court has come to a conclusion on this initial characterisation, it does not necessarily mean it will apply its domestic law to resolve the issue of what the staker and the staking intermediary did actually achieve in law. It may well be that the choice-of-law rules of the bankruptcy court points to another law as governing the particular matter in dispute. Thus, the importance of the *lex fori concursus* in insolvency matters does not mean that a bankruptcy court will never apply a foreign law. Quite to the contrary, it may do so quite routinely to determine whether property claims to an asset in the insolvency estate is properly made out under the relevant applicable law.

The exact outcomes of this process of identifying the applicable law is further complicated by the diversity of conflict-of-laws rules around the world. For example, the conflict rules of State X might state that, in the absence of choice between the parties, contractual matters are governed by the law with the closest connection with the contract, whereas the conflict rules of State Y state that, in the absence of a choice, contractual matters are governed by the law of the place where the contract was concluded. Although there is a degree of harmonisation in the EU, the general rule remains that each jurisdiction around the world has its own conflict-of-laws rules. Even

¹¹ See e.g. Antonio Leandro, 'Insolvency, applicable law' in Jürgen Basedow et al. (eds) *Encyclopedia of Private International Law* (Edward Elgar 2017) 931 (932).

within a single political nation state such as the UK or the US, each legal jurisdiction within these states follows its own domestic approach to the conflict of laws.¹² Accordingly, the question of the law applicable to our staker's rights will depend, to a large extent, on where our staking intermediary's insolvency is opened and administered, as bankruptcy courts will always and without exception follow the conflict-of-laws rules of the forum.

Accordingly, to answer the seemingly simple question of a staker's rights on its intermediary's insolvency, we start with a review of the various different crypto investment strategies currently seen in the market (B), which will then be subjected to a legal analysis to arrive at possibilities for legal characterisation (C). In the absence of any firm legal guidance at present on characterisation, our conclusions will be used to provide a tentative basis for determining the governing national law, at least in the abstract (D), before we draw some broader conclusions (E).

B. Empirical Distinctions

The ways in which cryptoassets may be used as an investment asset is limited only by the – often overly optimistic – creativity and ingenuity of those decentralized financiers at the forefront of the cryptoasset and Decentralized Finance (DeFi)¹³ spheres. Although there is no consistent use of terms, distinguishing between the various practices as a matter of empirical substance is crucial, as this ultimately feeds into the set of facts a court will be asked to consider and from which legal conclusions will be drawn. In this respect, there is, at least *prima facie*, a huge difference between, for example, an outright transfer of the staked asset from the staker to intermediary to be applied to the intermediary's investment strategy, and the exercise by the intermediary of certain rights associated with the staked asset without any transfer of the asset itself.

¹² For an overview, see Eugene F. Scoles et al., *Conflict of Laws* (6th edn, West Group 2018); John F. Coyle et al., *Choice of Law in the American Courts in 2021: Thirty-fifth Annual Survey by John F. Coyle, William S. Dodge, Aaron D. Simowitz* (2022), <https://dx.doi.org/10.2139/ssrn.4022722> (last visited 23 December 2022).

¹³ Decentralized Finance (DeFi) can be defined as follows: DeFi is a competitive, contestable, composable and non-custodial financial ecosystem build on technology that does not require a central organisation to operate and that has no safety net. See *Raphael Auer et al., 'The Technology of Decentralized Finance (DeFi)'* (2023) BIS Working Papers No 1066, 3.

Accordingly, this section sets out some of the various practices seen in the crypto investment context.

1. Staking

Staking in a more narrow, technical sense refers to a process peculiar to ‘proof of stake’ consensus mechanisms.¹⁴ Blockchain networks function on the basis of consensus¹⁵ and such consensus is achieved by one participant in the network – the so-called “validator” or “validating node” – collating proposed transfers broadcast to the network into a proposed new block, then propagating that block for acceptance by the other participants, or nodes, as the next block to be added to the chain. In most networks, any node can act as a validator, but as this process is open to abuse by malicious nodes, any validating node seeking to propagate a new block on the blockchain is required to ‘prove’ its trustworthiness. In the original ‘proof of work’ consensus mechanism, proving trustworthiness is done by solving a mathematical puzzle.¹⁶ By contrast, in the newer ‘proof of stake’ mechanism, the validating node must both (i) prove it holds a ‘stake’ in the network, i.e. a quantity of the blockchain’s native crypto-asset;¹⁷ and (ii) be willing to forfeit that stake in the event that it has been

¹⁴ These are increasingly being adopted by blockchain networks to replace the energy-intensive ‘proof of work’ mechanism. See e.g. the efforts to replace the proof-of-work mechanism of the Ethereum network by a proof-of-stake consensus mechanism, which led on 15 September 2022 to what became known as The Merge; available at: <<https://ethereum.org/en/upgrades/merge/>> (last visited 21 December 2022). On the debate about the sustainability of the proof-of-work mechanism, see i.a. ADAN, “Ban of Proof-of-Work protocols: wrong answer to real global environmental challenges”, 28 February 2022, available at: <<https://adan.eu/en/position/pow-bitcoin-ban-eu-inappropriate-answer>> (last visited 21 December 2022); see also *Felix Irresberger et al.*, ‘The public blockchain ecosystem: An empirical analysis’ (2021) NYU Stern School of Business, available at SSRN: <<https://ssrn.com/abstract=3592849>> (last visited 21 December 2022); On different consensus protocols see *Raphael Auer et al.*, ‘The Technology of Decentralized Finance (DeFi)’ (2023) BIS Working Papers No 1066, 8; *Shijie Zhang and Jong-Hyouk Lee*, ‘Analysis of the main consensus protocols of blockchain’ (2020) 6(2) ICT express, 93-97.

¹⁵ A very good introduction to blockchain technology is found in Arvind Narayanan et al., *Bitcoin and Cryptocurrency Technologies. A Comprehensive Introduction* (Princeton University Press, 2016); for a slightly older introduction, see Andreas Antonopoulos, *Mastering Bitcoin: Unlocking Digital Cryptocurrencies* (O’Reilly 2014); see also Aaron Wright and Primavera De Filippi, ‘Decentralized Blockchain Technology and the Rise of Lex Cryptographia’ (2017); Philipp Maume and Lena Maute, *Rechtshandbuch Kryptowerte* (C.H.Beck/Vahlen 2020), 1-32; Sebastian Omlor and Mathias Link, *Kryptowährungen und Token* (Deutscher Fachverlag 2021), 13-17, 72-109; Iris H-Y Chiu, *Regulating the Crypto Economy* (Hart 2021) 1-44; Carol Goforth, *Regulation of Cryptotransaction* (West Academic Publishing 2020) 1-10; Colleen Baker and Kevin Werbach in Jelena Madir, *Fintech Law and Regulation* (Edward Elgar 2021) 148-164.

¹⁶ See Narayanan et al., 131-164; see also Daniel Stabile, Kimberly Prior and Andrew Hinkes, *Digital Assets and Blockchain Technology* (Edward Elgar 2020) 20-22; *Raphael Auer et al.*, ‘The Technology of Decentralized Finance (DeFi)’ (2023) BIS Working Papers No 1066, 8.

¹⁷ See Daniel Stabile, Kimberly Prior and Andrew Hinkes, *Digital Assets and Blockchain Technology* (Edward Elgar 2020) 22.

deemed by the other nodes as fraudulent or as having otherwise acted in bad faith¹⁸ in a process referred to as ‘slashing’.¹⁹

Crypto-assets thus offered up as ‘proof’ of a validating node’s ‘stake’ in the next block are known as ‘staked,’ and the risk of losing staked assets through the sanction of slashing works to deter any validating node from impairing the smooth functioning of a blockchain network. Backed by a slashing mechanism, staking thus ensures the integrity of any new block added to the blockchain.

For its validation efforts, the validating node earns rewards in the form of the transaction fees paid by users of the network and/or crypto-assets newly created with each propagated block. Nodes, therefore, compete for its proposed blocks to be accepted onto the blockchain. The exact selection process varies across proof-of-stake algorithms, however. Two models of staking are worth distinguishing: direct staking and staking by delegation.

a) Direct Staking

A node can directly stake crypto-assets it holds to the network by participating in the ‘proof of stake’ consensus mechanism. In this case, it is the node itself that suffers the loss if its stake is ‘slashed’ due to fraudulent transactions or inactivity.

b) Staking via an Intermediary (‘Delegation’)

Staking can also be done via an intermediary. In practice, this is the most common form of staking.²⁰ Many crypto intermediaries – notably crypto exchanges – offer

¹⁸ See *infra* Section C.1.

¹⁹ Such slashing takes place, for example, with Polkadot (<https://wiki.polkadot.network/docs/learn-staking#slashing>), Tezos (<https://medium.com/everstake/how-does-slashing-work-in-tezos-and-why-is-it-important-to-delegate-only-to-reliable-bakers-like-a6c931e93c56>) and, in future, probably also with Ethereum (<https://www.bloxstaking.com/blog/ethereum-2-0/understanding-eth2-slashing-preventative-measures/>). In contrast, Avalanche (<https://www.avax.network/validators>) and Cardano (<https://cardanians.io/en/why-cardano-does-not-need-slashing-152>) only refuse to reward the staker in such cases, but the crypto-assets themselves are not withdrawn. Solana (<https://docs.solana.com/de/proposals/slashing>) also provides for such a mechanism in principle but it has not yet been activated; see on the functioning of slashing exhaustively Fanti, Kogan and Viswanath, ‘Economics of Proof-of-Stake Payment Systems’, p. 11 et seq., 21, available at: <https://cbade.hkbu.edu.hk/wp-content/uploads/2021/05/20210520_KOGAN_2.pdf> (last visited 23 January 2023).

²⁰ Currently the following platform offer staking by delegation: *Coinbase*, see Coinbase User Agreement (January 2023), Appendix 6, available at: <www.coinbase.com/legal/user_agreement/ireland_germany> (last visited 23 January 2023); *Binance*, see Binance Terms of Use (January 2023), 9. Staking Programs, available at: <www.binance.com/en/terms> (last visited 23 January 2023); *Kraken*, Terms of Service, Annex C and Annex D, available at: <www.kraken.com/legal> (last visited 23 January 2023); *Bitpanda*, Allgemeine Geschäftsbedingungen, Version 11.0.1 (November 2021), Annex II A and Annex II B, available at:

staking services to their accountholders. They can do so by either acting as a validator or by mediating the staking of the crypto-assets to a third party who will act as a validator.²¹ There are important commercial incentives for doing so: broadly speaking, the greater the quantity of crypto-asset a validating node can stake, the more returns it can generate from the validation process. Hence, it makes sense to collect crypto-assets from those willing to stake, use them in the validation process and share the rewards with those that have contributed to the staking of the intermediary.

Staking via an intermediary is often called 'delegation', even though the terminology varies. Two types must be distinguished.

In the first, crypto-assets are contributed to the pool of assets staked by the validator itself or by a third party acting as the validating node.

In the second type of 'delegation', crypto-assets are not transferred themselves, but rather, a 'right' to propagate the next block is transferred. This method of staking via delegation is associated with particular types of blockchain networks,²² where each unit of the blockchain's native crypto-asset inherently carries such right. The right to propagate the next block is allocated at random but many holders of the native cryptocurrency are uninterested in exercising the right. It therefore 'delegates' this right to a node able and wishing to exercise it, usually in return for a fee. Crucially, only the 'protocol right to propagate the next block' associated with the crypto-asset is delegated, not the asset itself.

<www.bitpanda.com/de/legal/bitpanda-general-terms-conditions> (last visited 23 January 2023); and many more.

²¹ See *Coinbase*, see Coinbase User Agreement (January 2023), Appendix 4, 3. Staking Services, available at: <www.coinbase.com/legal/user_agreement/ireland_germany> (last visited 23 January 2023): ("[...] you may be given the option to "stake" these assets in a third party proof of stake network via staking services provided by Coinbase or an affiliate or a third party"); *Kraken*, Terms of Service, Annex C and Annex D, available at: <www.kraken.com/legal> (last visited 23 January 2023): ("Payward Trading may perform any or all of the Services described in this Addendum directly or through one or more service provider(s)."); *Bitpanda*, Allgemeine Geschäftsbedingungen, Version 11.0.1 (November 2021), Annex II A, 1.2., available at: <www.bitpanda.com/de/legal/bitpanda-general-terms-conditions> (last visited 23 January 2023): ("Der Staking Dienst wird in Zusammenarbeit mit Drittanbietern von Staking Diensten angeboten."); and many more.

²² Tezos Whitepaper, 'Tezos – a self-amending crypto-ledger' (2014), 10-14, available at: <<https://tezos.com/whitepaper.pdf>> (last accessed 23 January 2023); Amy Held, 'Baking, staking, Tezos and trusts: crypto sale and repurchase transactions analysed by the High Court' (2022) *Butterworths Journal of International Banking and Financial Law*, 96 (96); Seyed Mojtaba Hosseini Bamakan, Amirhossein Motavali and Alireza Babaei Bondarti, 'A survey of blockchain consensus algorithms performance evaluation criteria' (2020) 154, *Expert Systems With Applications*, 5; Nicola Dimitri, 'Liquid Proof-of-Stake in Tezos: An Economic Analysis' (2022) 13(12) *Information*, 556 (556-557).

2. *Yield Farming*

The term ‘yield farming’ is associated with DeFi; in particular, with decentralized crypto exchanges (DEXs). DEXs are platforms on which crypto-assets can directly be exchanged without the intervention of any intermediary.²³ Although the term is used in various ways, in practice, ‘yield farming’ is, for the purposes of this paper, used to denote the process whereby the DEX incentivises its users to stabilise the value of certain crypto-assets – usually the DEX’s own native token/cryptocurrency – by refraining from trading in those assets.

This commercial objective of reducing the volatility of the relevant crypto-asset is effected by the user committing assets of that type to a smart contract, with returns paid by the DEX in exchange. Assets committed in yield farming operations are held passively in the smart contract and are not traded on the exchange or used for some other purpose. The user is given a receipt to acknowledge the deposit, usually in token form and is usually permitted to withdraw the crypto-assets at will.²⁴

3. *Liquidity Mining*

Liquidity mining also involves a DEX, but differs in that a crypto-asset holder will provide a pair of crypto-assets to a DEX, and will receive in return crypto-assets issued by the DEX as ‘interest’.

To understand the process of liquidity mining, it is important to understand a little further how DEXs operate. In general, trading of crypto-assets on a DEX takes place via “automated market makers” (AMM), which are smart contracts that provide much needed liquidity to the market.²⁵ In order to fulfil their task, AMM themselves need liquidity in the form of crypto-assets. This is necessary for the DEX to function

²³ See *Raphael Auer et al.*, ‘The Technology of Decentralized Finance (DeFi)’ (2023) BIS Working Papers No 1066, 18-20.

²⁴ E.g. SYRUP Tokens at PancakeSwap, <<https://docs.pancakeswap.finance/products/syrup-pool/syrup-pool-guide/syrup-pool-faq>> (last visited 23 January 2023), xSUSHI at SushiSwap, <<https://docs.sushi.com/docs/Products/Yield%20Farming/The%20SushiBar>> (last visited 23 January 2023), rJOE, sJOE and veJOE at Trader Joe, <<https://help.traderjoexyz.com/en/trader-joe/staking>> (last visited 23 January 2023), veCRV at Curve Finance, <<https://resources.curve.fi/crv-token/staking-your-crv>> (last visited 23 January 2023).

²⁵ On AMMs see *Alfred Lehar and Christine A. Parlour*, ‘Decentralized Exchanges’ (2021), available at: <<https://ssrn.com/abstract=3905316>> (last visited 24 January 2023); *Lindsay X Lin*, ‘Deconstructing Decentralized Exchanges’ (2019) Stan. J. Blockchain L. & Pol’y; *Raphael Auer et al.*, ‘The Technology of Decentralized Finance (DeFi)’ (2023) BIS Working Papers No 1066, 11-13.

continuously and smoothly, which is particularly important for the trading of digital assets with a low trading volume.

On a DEX, the liquidity will be provided by a so-called Liquidity Provider (LP), i.e. a holder of crypto-assets, by putting them into a so-called Liquidity Pool. Technically, this will be implemented by the LP exchanging a pair of – at least two – crypto-assets against a LP token. As a rule, the possible pairs are predefined and presuppose a certain ratio between the crypto-assets provided. The LP token is not a mere receipt of deposit, but represents the share of the liquidity pool that the contributed crypto-assets make up.²⁶

The process of liquidity mining is, thus, quite similar to yield farming. However, it is distinct from yield farming operations in several key ways.

First, the commercial objective of the offeror of such transaction differs. Whereas yield farming aims at reducing the circulation of certain crypto-assets, liquidity mining serves to provide additional liquidity of a certain type of assets. Unlike deposits of crypto-assets provided in yield farming operations, contributions in liquidity mining do not remain passive, but are actively used in the market by the DEX. The DEX uses the liquidity so that market participants can exchange the respective crypto-assets in the liquidity pool against each other. The exact combinations of crypto-assets to be contributed are usually predefined by the DEX, but there generally are no caps as such on the number of crypto-assets a DEX will accept into a Liquidity Pool.

Second, Liquidity Providers may 'cash out' their share of the Liquidity Pool for the contributed crypto-assets at any time by redeeming the LP Token. There is, however, no guarantee that the Liquidity Provider will receive the same amount of each crypto asset as they had originally contributed; all depends on the composition of the Liquidity Pool at the time of redemption. For this reason, and because of the fluctuating value of crypto-assets, LPs endure the risk of impermanent loss (or divergence loss, as it is sometimes called), which occurs when one provides liquidity to a pool that contains more than one asset and then the market price of at least one

²⁶ *Raphael Auer et al.*, 'The Technology of Decentralized Finance (DeFi)' (2023) BIS Working Papers No 1066, 10, 12: ("ETH/USDC LP is an example for an equity token, representing claims on shares of underlying assets, in this specific case being a claim on the amount of ETH and USDC deposited as liquidity provision (LP) in the Uniswap Dex."). In V3 version of the Uniswap Dex implemented an NFT representing fractional ownership of the protocol liquidity. See <<https://uniswap.org/blog/uniswap-v3>>, (last visited 24 January 2023).

of the assets in the pool changes.²⁷ An impermanent loss thus derives from the fact that the returned assets upon redemption might be of a lesser total value than if they had simply been held onto and not used for liquidity mining.

4. *Crypto Lending*

Crypto lending enables users to lend and borrow crypto-assets for a fee or interest. These transactions can be offered by centralised²⁸ or decentralized²⁹ platforms, or by other operators. Under such a lending model, the holder transfers crypto-assets to another address for the payment of remuneration. The recipient can then freely transfer or pledge the crypto-assets until it receives a withdrawal order from the address depositing the crypto-assets. This withdrawal order can often be placed at any time. Higher interest rates may, however, be achieved if crypto-assets are invested for a fixed period of time.

Decentralized platforms offer lending transactions via lending pools.³⁰ Crypto-assets are contributed to a pool to be lent onwards to third parties against collateral and interests. This is realised through a smart contract, through which a crypto-asset is transferred to a DeFi platform in exchange for a corresponding crypto-asset issued by the platform.³¹ Each borrowed crypto-asset thus matches its token counterpart of the DeFi platform. The token issued by the DeFi platform can – possibly after a certain

²⁷ Doncho Karaivanov, 'Impermanent Loss Explained With Examples & Math' (2021) available at: <<https://chainbulletin.com/impermanent-loss-explained-with-examples-math/>> (last visited 9 April 2022).

²⁸ For example see Binance, <www.binance.com/en/savings#lending-demandDeposits>; Bitfinex, <www.bitfinex.com/lending-products-start/>; BlockFi, <www.blockfi.com>; Circle, <www.circle.com/en/products/yield>; Crypto.com, <<https://crypto.com/eea/earn>>; Gemini, <www.gemini.com/earn> (all last visited 9 April 2022).

²⁹ For example see Aava, <<https://aave.com/>>; Compound <<https://compound.finance/>>; MakerDao, <<https://makerdao.world/en/learn/vaults/>> (all last visited 9 April 2022); see also *Raphael Auer et al.*, 'The Technology of Decentralized Finance (DeFi)' (2023) BIS Working Papers No 1066, 13-15; Daniel Perez et al., 'Liquidations: Defi on a Knife-Edge' In International Conference on Financial Cryptography and Data Security (Springer 2021) 457-476.

³⁰ *Raphael Auer et al.*, 'The Technology of Decentralized Finance (DeFi)' (2023) BIS Working Papers No 1066, 13-15, (highlighting that an "essential difference to loans issued by traditional financial institutions is that interest rates are set automatically"); further see Jiahua Xu and Nikhil Vadgama, 'From Banks to Defi: the Evolution of the Lending Market' in *Enabling the Internet of Value* (Springer 2022), 53-66.

³¹ An example is the cToken issued by the platform Compound against the deposit of an eligible type of crypto-asset, see *Robert Leshner and Geoffrey Hayes*, 'Compound: The Money Market Protocol' (2019), 3-4, available at: <<https://compound.finance/documents/Compound.Whitepaper.pdf>> (last visited 23 January 2023); another example would be Aave with its aToken, see Aave Whitepaper, 8-16, available at: <https://github.com/aave/aave-protocol/blob/master/docs/Aave_Protocol_Whitepaper_v1_0.pdf> (last visited 23 January 2023).

period of time – be redeemed for the crypto-asset originally provided, with an additional remuneration. This makes crypto lending look very similar to liquidity mining. The difference, however, is that in crypto-lending, assets are not provided in pre-defined pairs. Also, the lent assets can be used for a wider range of purposes.

5. Summary

As has been demonstrated by this brief survey, whilst ‘staking’ is generally used in a flexible way by market participants, use of crypto-assets to generate passive income encompasses a wide range of practices. For reasons that will become apparent, we consider that, when trying to ascertain the rights of a staker on its intermediary’s insolvency, it is imperative to distinguish between various ‘staking’ models.

C. Legal Characterisation

As noted in Section A, characterization is a matter undertaken by the forum as a preliminary step in determining the law applicable to any issue in dispute before it. In the case of the rights of a staker in the event of its intermediary’s insolvency, the most pressing issue in dispute will usually be whether the staker has a proprietary interest in an asset that has come to fall within the insolvent estate, or whether his rights are merely personal. Though bankruptcy proceedings are generally governed by the law of the bankruptcy court (the *lex fori concursus*), this is not the case for proprietary interests. The law applicable to them will be determined separately, using classic conflict-of-laws rules, to protect vested rights that have been acquired before the opening of the insolvency proceedings. Crucially, classic conflict of laws provides varying rules for different assets and legal relationships.³² It is therefore indispensable to first characterize the legal operation in question before one can identify the law applicable to them.

In order to characterize the legal nature of the operation, the bankruptcy court will consider the circumstances in which the staking intermediary acquired the asset: i.e., the legal basis upon which the staker conceded control over the asset in favour

³² See *infra* Section D.

of the intermediary (which is why the asset now has fallen into the intermediary's insolvency estate). Only then will it be possible to ascertain the legal consequences flowing from the transfer of control. Accordingly, we consider now some of the ways in which the practices in Section B could be characterized as a matter of law, and the legal consequences that follow. To do so, we rely to a considerable extent on the express terms and conditions or 'staking agreement' pursuant to which the staker conceded control over the asset in the first place. We are mindful of the fact that these agreements both (i) are not conclusive as to what the parties achieved in law, and (ii) may well not reflect what actually happens at protocol level from a technological point of view. Nevertheless, an express agreement between the parties will be a significant factor to be considered by the court as they will *prima facie* provide some indication of what the parties had intended to do, objectively assessed.

In order to take account of as many legal systems as possible, we do not commit ourselves to a single generally valid classification, but rather propose a number of possible legal categories. While we will cite literature from various countries, we do not mean to refer exclusively to any particular national legal system, but rather to a "transnational" notion.

Finally, the categories we use are those that apply to private transactions. We are not concerned with regulatory law, such as characterizations of crypto-assets as "securities" or "commodities" by the SEC or the CFTC. These are exclusively valid in the realm of supervision and regulation. They do not determine the rights and obligations between private individuals, such as crypto holders and crypto exchanges, and do not apply in litigation between them.

1. Secured Transaction

When looking at staking (in the technical sense) from a legal perspective, "secured transactions" immediately comes to mind.³³ To create a security interest, the owner of an asset, the "security provider", usually gives up possession or control over the asset.³⁴ The security right is "perfected" when the other party, termed "secured party",

³³ See Art 9 of the UCC; see also UNCITRAL Model Law on Secured Transactions (2016).

³⁴ See for the U.S.: UCC 9-203(b); Steven L. Harris and Charles W. Mooney, *Security Interests in Personal Property* (5th edn Foundation Press 2011), p. 91 (mentioning three conditions UCC 9-203(b) sets forth to enforceability, and thus attachment: First, value must be given; second, the debtor must

obtains possession over the asset or when a financing statement is filed.³⁵ The security interest or collateral right is a limited property right, protecting the security taker in case the security provider fails to perform his obligations, e.g. repaying a loan. The security right allows the security taker to obtain control over the asset if he does not yet have it (i.e. in case a financing statement has been filed). He may then sell the asset and apply the net proceeds of the sale to the secured obligation.³⁶

Ignoring for one moment the differences between tangible and intangible assets, the simple form of staking bears some striking resemblances with a secured transaction: Typically, the asset will be locked during the time it is staked, meaning that the staker is prevented from exercising control over them.³⁷ During the whole process, the asset can be accessed only by the staker; the providers of staking services are at pains to underline that the staker retains "ownership" despite the staking.³⁸ Finally, if the staker or the validator has violated the rules of the network, e.g. by engaging in malicious validation or by refusing to validate, the assets may be "slashed", which essentially means that they will be transferred to the network.

All of these points make staking seem similar to a secured transaction. A further indication supporting this characterization is the use of the word "pledge" by some staking agreements;³⁹ a pledge is a typical secured transaction. There can hardly be

have rights in the collateral, and this the debtor must agree that a security interest will attach and either the collateral must be in the secured party's possession or control or the debtor must have signed a security agreement); *Raphael Auer et al.*, 'The Technology of Decentralized Finance (DeFi)' (2023) BIS Working Papers No 1066, 4 ("Customers that interact with such platforms [...] give up custody of their assets").

³⁵ UCC 9-201; Harris and Mooney, id., p. 92-93 (calling UCC 9-201 a "baseline rule", which is generally understood to mean that an attached security interest will be senior to conflicting claims unless the UCC provides otherwise).

³⁶ Harris and Mooney, id., p. 617 ("Taking possession of collateral following a debtor's default only begins the process of enforcing a security interest in collateral. A secured party in possession of tangible collateral following a default typically wishes to sell the collateral and then to apply the net proceeds of the sale to the secured obligation").

³⁷ See e.g. Coinbase User Agreement, March 16, 2022, sub 3 ("Staking Services") 1.2 ("Some Digital Asset networks require that a certain amount of staked assets be locked (restricted from sale or transfer) for a certain period of time while staking."). See also, id., 1.5 (ii) ("Lockup Period") ("If you choose to stake your ETH, your ETH will be pledged for staking and will become locked on the Ethereum protocol until Phase 1.5 of the Ethereum network upgrade is completed.")

³⁸ See Kraken Terms of Service March 22, 2022, Annex C 1.2. ("You retain ownership of the Supported Tokens and such Supported Tokens shall remain property of you when staked under the terms of this Addendum."), Annex D 2.d ("You remain the lawful and beneficial owner of Staked Assets..."); Bitpanda Terms of Service, Annex 7.2 ("Bitpanda Clients are the beneficial owners of Stakes Assets at all times, ...")

³⁹ See Coinbase User Agreement, March 16, 2022, 1.5 (ii) ("Lockup Period") ("If you choose to stake your ETH, your ETH will be pledged for staking...").

a stronger indication of the parties' intentions than the use of this word, which is understandable even to laymen.

Yet, there are also some glaring differences between staking and classic secured transactions. First, the staker can usually "unstake" the assets at any time and thereby immediately terminate the staking agreement; in this case, the assets will be returned to her subject to the non-occurrence of a slashing event.⁴⁰ Second, the security interest does not secure the repayment of a loan, but rather the proper performance of the validation functions on the network; in other words, the assets will be taken away as a punishment for not behaving properly on the network rather than for not fulfilling a monetary obligation. It does not serve to mitigate credit risk but gives incentives for 'good' behaviour.⁴¹

One may reasonably debate whether these two particularities exclude to characterize staking as a secured transaction. The first point, it is suggested, does not necessarily justify such an exclusion, as a security provider and security taker are free to determine the length of their agreement and can also include a right to terminate it at any time. The second particularity is perhaps not as great if one considers that also collateral incentivizes 'good' behaviour in the sense of a loan. Still it must be admitted that there is no underlying obligation to repay a loan and the associated credit risk mitigation. However, it is not completely excluded that a secured transaction may serve to enforce some other obligation; just think of bail that is deposited with a court or the police or security for cost that is provided in a litigation. In these cases, the asset may be lost if some other event than the non-repayment of the loan occurs. Still, one may speak of a "secured transaction", at least in a loose sense of the word.

Other types of transaction that are common in the field of generating revenue from crypto-assets do not bear this similarity to secured transactions. While the token holder also relinquishes possession or control over the asset in the case of yield farming, liquidity mining, and crypto lending, these cases lack a legal obligation to be collateralized. We therefore consider it highly unlikely that these transaction types will be considered to fall into the category of secured transactions.

⁴⁰ See Bitpanda Annex 2.7 ("No lock-up.") ("The Staked Asset can be unstaked at any time with immediate effect. In this case the Bitpanda Client receives his Staked Assets, subject to the non-realization of the Slashing Risk outlined in point 7, back in their Wallet and all Rewards accrued on such Staked Assets until the relevant Final Offer Click Unstaking.")

⁴¹ See *James Burnie, Meghan Millward and Matthew Kimber, 'What's at Stake? The Legal Treatment of Staking' (2022) Butterworth's Journal of International Banking and Financial Law 594, 596.*

The effect of characterizing an operation as a “secured transaction” depends on the type of the secured transaction. In case of a “pledge” or “lien”, the operation would not touch upon the initial distribution of ownership. There would merely be the creation of a new right *in rem*. The insolvency risks of the person providing the security right would thus be limited. If, on the other hand, the secured transaction results in an “outright transfer”, then the collateral provider would bear the full risk of insolvency of the collateral taker. Which of these two types of characterizations will be retained in a particular case is not entirely sure and will depend on the applicable terms and conditions. Yet our conjecture is that without any contrary indications, it is more likely that a court will lean in favour of a more restricted transfer, i.e. the creation of a pledge or lien. That is because this category strikes an appropriate balance between the interests of the staker and those of his counterparty: the staker does not lose his rights completely, while the counterparty gets as many rights as it needs for its particular purpose, i.e. slashing.

2. *Loan*

The legal concept of a loan is, perhaps, particularly apt for crypto lending. Broadly defined, a loan is a transaction whereby one party – the lender – makes funds available to another party – the borrower – against an obligation to repay the nominal amount, with or without interest. Crypto lending seems consonant with this definition, because it implies the full title transfer of the relevant assets to another party with an obligation to return goods of the same quality and quantity plus remuneration.

At least at first blush, lending crypto-assets to a DEX may therefore be legally characterised as a loan. Loans in law, however, require that several other criteria be met, and whilst each jurisdiction differs, some broad principles are common to all.⁴²

The first criterion is that the economic purpose of a loan agreement must be to provide capital to the borrower. The debtor is permitted to use the capital for a certain period of time, e.g. to make investments or grant loans itself. There must not be any restrictions on how the crypto-assets are used by the platform. From the lender’s perspective, this means he must give up the right to claim back the exact same assets

⁴² Matthias Lehmann, ‘Bonds and loans’, in Jürgen Basedow et al. (eds) *Encyclopedia of Private International Law* (Edward Elgar 2017) 215-219.

in specie; he must, therefore, be willing to give up his title to the assets against the promise to receive goods of the same quantity and quality.

The second condition is that the borrower must be obliged to repay the funds. This is generally the case. Crypto lending services provide for the reimbursement of the deployed funds, generally including interest.⁴³ This is not surprising – after all, hardly anyone is willing to donate their assets. Typically, the parties to crypto lending stipulate that the repayment is made upon the lender's request at any time. Traditionally, repayment of loans are, however, only accelerated under specific and limitative conditions. Of course, the determination of the due date is generally in the hands of the parties, and loan agreements come in different kinds; bullet loans, for instance, are loans in which the principal is repaid in full at the end of the loan term, but here again repayment is made by the borrower and not upon simple and unconditioned recall by the lender. This notwithstanding, the qualification of a loan might prove particularly apt for crypto lending schemes.

While the characterization as a loan might be particularly apt for crypto lending, other transactions are less likely to fall into this category. Whereas in staking, yield farming or liquidity mining the initial token holder may also result in an outright transfer of funds, their economic purpose is not to provide capital to the borrower. Staking serves to secure a position of a network. Yield farming aims at reducing the amount of tokens in circulation. Liquidity mining is perhaps closest to a loan, yet the liquidity miner receives as counter-performance an interest in the LP, which is unusual for a lender. We therefore consider it rather unlikely that any of these other three operations will be qualified as a loan in any jurisdiction.

The key point of the characterization as a loan is its repercussion on the distribution of risks, especially insolvency risks. Like a loan of fiat currencies or other movable assets, the loan entails the transfer of crypto-assets from the lender to the borrower. The initial token holder is losing any right she had in the asset and will have only a claim – i.e. an obligation – against the borrower. In the insolvency of the borrower, the initial token holder will therefore rank as unsecured and may then be left only with a part (quota) of the amount he or she has lent or suffer a complete loss. This shows the importance of distinguishing crypto lending from staking.

⁴³ See e.g. Blockchain.com terms and conditions under “5.1 Passive Rewards”, available at: <<https://exchange.blockchain.com/legal/terms>> (last visited 23 January 2023).

3. *Deposit*

Alternatively, some practices could also be characterized as deposits, which has its origins Roman law. Remnants of the Roman concept of *depositum* survives in both modern common law and civilian jurisdiction alike, and has particular influence on the law of banking. Accordingly, the characterisation seems particularly apt, given the comparisons that have been drawn between crypto-lending to DEXs with depositing money with a bank.⁴⁴ To test the validity of this comparison, it is necessary to distinguish between two types of deposits.

The first type of deposit – called in Latin *depositum regulare* – serves the commercial objective of safekeeping the assets in the interests of the depositor. A typical use case would be a deposit box in the vault of a bank of which the depositor has exclusive use to deposit his assets. Under this model, the assets deposits remain in the vault, and are recovered *in specie* upon a withdrawal by the depositor, who does not lose his right of ownership in the assets deposited.

However, the business of banking has long discarded this model of deposit; adopting, instead, what is known in many civilian jurisdictions as *depositum irregulare*.⁴⁵ This follows fractional reserve banking, under which a bank uses the assets of the depositors, for instance to provide credit to third parties. In this model, the custodian acquires ownership of the asset held in custody and has all the rights of the owner to dispose of the fungible assets held in custody; his obligation to the depositor is simply to return goods of the same quantity and quality.⁴⁶ This model is described by the following quote by Lord Cottenham in *Foley v. Hill*:

The money paid into the banker's, is money known by the principal to be placed there for the purpose of being under the control of the banker; it is then the banker's money; he is known

⁴⁴ See Sorkin et al., 'The New "Shadow" Banks', The New York Times, 8 September 2021, <www.nytimes.com/2021/09/08/business/dealbook/crypto-bitcoin-regulation.html> (last visited 23 January 2023).

⁴⁵ See Sec 700 German Civil Code; Art 481 Swiss Code of Obligations; Art 1782 of the Italian Civil Code in connection with Art 1834 of the Italian Civil Code; Art 1992 of the Latvian Civil Code; see Art. 1932, al. 2 French Civil Code and the relevant case law, viz. Court of Cassation, Civ. 1, 29 Nov. 1983, no. 82-14.490 and Court of Cassation, Com., 9 July 1979, no. 78-12.988.

⁴⁶ *Timo Fest* in Münchener Kommentar zum HGB vol 6 (4th edn, CH Beck 2019), Einlagengeschäft para 193-194; *Martin Henssler* in Münchener Kommentar zum BGB, vol 6 (9th edn, CH Beck 2023) § 700 BGB para 3.

*to deal with it as his own; he makes what profit he can, which profit he retains to himself [...]*⁴⁷

The characterization as an irregular deposit may seem appropriate for some of the operations analyzed where the following two conditions are met: first, the transferor can recall the assets at any time, and second, the recipient does not have to return the same assets he received. This is the case in particular for yield farming and liquidity mining, but also for crypto lending. It is thus not excluded that a court may characterize any of these three operations as an irregular deposit.

However, we are of the opinion that such characterization would not correspond to the commercial purpose of the operations and the intentions of the parties that underpin them. Typically, the underlying crypto assets are held with a custodian. Since they are already in custody, there is no reason for an additional deposit with the same or another intermediary. Instead, the commercial reasoning behind the operations described is to obtain profits, and this not only on the side of the recipient of the asset, but also on the part of the transferor. However, it is not inconceivable that for certain actors the purpose of safekeeping is of overriding importance. In this case, a characterization as a deposit may be apposite.

While the answer whether a platform is taking deposits ultimately depends on the parties' intentions and on how the services offered are structured, the risk analysis remains the same as in the case of crypto lending: the initial holder of the crypto-asset bears the counterparty risk, i.e. the risk that the other party to the contract does not perform (see above section C(2)). It is therefore not significant for the distribution of insolvency risks whether one characterizes crypto lending as a loan or as an irregular deposit.

4. Partnership

Partnerships exist in many jurisdictions in different shapes and forms. At their basis is an association of two or more persons to carry on as co-owners a business for profit.⁴⁸

⁴⁷ *Foley v Hill* (1848) 2 HLC 28, 9 ER 1002; see also Peter Ellinger, Eva Lomnicka and Richard Hooley, *Modern Banking Law* (3rd edn, Oxford University Press 2002), 93-94; further see *Joachimson v. Swiss Bank Corporation*, [1921] 3 KB 110; *Busher, Clerk of Courts, v. Fulton, Supt. of Banks*, 128 Ohio St. 485, 191 N.E. 752.

⁴⁸ Section 1 UK 1890 Partnership Act. In the United-States, partnership legislation belongs to the states' sphere of competence but is harmonised via the Uniform Law Commission (ULC) Uniform Partnership Act.

So-called general partnerships are particularly attractive as they do not require registration with a company registrar. Similar regimes are to a certain extent also known to civil law jurisdictions.⁴⁹

Absent any registration or incorporation, crypto holders staking their crypto-assets may at first sight qualify as partners of a general partnership, where all other crypto holders contributing assets are partners. As described in the empirical section,⁵⁰ all stakers indeed undertake to add and validate new blocks of a blockchain with the common goal of processing and timestamping transactions or support validators by contributing assets to the validation process in exchange for rewards. The prospect of profit may be seen as a clear *prima facie* evidence of a partnership. Stakers also submit and agree upon the proof-of-stake mechanism being a kind of randomised scheme of distribution of profits.⁵¹ The characterisation as a general partnership is particularly relevant when stakers pool together their crypto-assets for the purpose of staking, as they clearly partner up (albeit not knowing each other personally) to profit from the addition of a new block. However, in some staking agreements the allocation of profits is randomised and upon the intermediary's discretion, which does not square well with the traditional concept of general partnerships.⁵² In this case, stakers also jointly bear the risks of, *inter alia*, slashing.

The partnership characterization also seems to stand independently of whether stakers act directly or indirectly via an agent, i.e. a validating node that will act on their behalf. In that sense, delegation is very much akin to a principal-agent relationship, irrespective of whether it entails an outright transfer of crypto-assets or a mere transfer

⁴⁹ E.g. Art 1871-1873 French Civil Code which sets out the legal regime of the unregistered “société en participation” that applies also to *de facto* partnerships (“sociétés créées de fait”); see Section 1175 of the Austrian Civil Code (ABGB) and § 705 German Civil Code (BGB), stating that if two or more persons pursue a joint (business) activity they form a civil-law partnership unless choosing another legal personality. The Austrian civil-law partnership, however, has no legal capacity and therefore cannot be a subject of its own rights and obligations.

⁵⁰ See *supra* Section B.

⁵¹ See e.g. *Blockchain.com*, Terms and Conditions, 5.2 Staking Services, available at: <<https://exchange.blockchain.com/legal/terms>> (last visited 23 January 2023).

⁵² *Coinbase*, User Agreement (January 2023), Appendix 6, 3. Rewards, available at: <www.coinbase.com/legal/user_agreement/ireland_germany> (last visited 23 January 2023); *Binance*, Binance Terms of Use (January 2023), 10. Binance POS Service Agreement, available at: <www.binance.com/en/terms> (last visited 23 January 2023); *Kraken*, Terms of Service, Annex C and Annex D, 1.3., available at: <www.kraken.com/legal> (last visited 23 January 2023); *Bitpanda*, Allgemeine Geschäftsbedingungen, Version 11.0.1 (November 2021), Annex II A, 3. Rewards, available at: <www.bitpanda.com/de/legal/bitpanda-general-terms-conditions> (last visited 23 January 2023) and many more.

of a 'staking rights'.⁵³ In such use case scenario, stakers (as principals) remain the partners.

This characterization seemingly also fits rather well liquidity mining and certain types of crypto lending, namely those operating with pools of crypto-assets. Both types of DeFi applications generally rely on at least two people collectively committing their crypto-assets to a common pool. In the case of liquidity mining, token holders provide liquidity to the market by transferring their crypto-assets to a common pool often in exchange of a token representing their share in the pool. In doing so, they pursue the common aim of generating returns. The same line of argumentation can be extended to yield farming.

Similar concepts may also be found in the case of crypto lending. Token holders provide capital to a third party (a DEX or another operator) to generate returns. Such third party may be a smart contract pool offered by a DEX. In this case, one might assume a partnership between all crypto-assets holders contributing to such lending pool.

The holders of the crypto-assets also bear a certain risk, namely that the borrower may not be able to repay the borrowed crypto assets.

There are, however, also a number of counter-arguments against the partnership characterization. First, it seems generally inapposite for mere bilateral relationships in which a crypto holder transfers outright its crypto-assets to another operator, without of common will to partner-up with the recipient for profit. This point is likely to point against the partnership characterization of most crypto lending operations.

Second, and even more importantly, the parties to the operations analyzed here will generally not be ready to assume the liability of the partners for all debt and obligations of the venture. Typically, crypto-holders that partner-up in a pool generally do not intend to assume any risk of liability beyond the loss of the asset(s) they committed and transferred.

A possible exception would be a case where pooled crypto-assets are used as collateral in leverage transactions for which liabilities might exceed the total asset value of the pool, but this scenario is very rare. In general, no staker, yield farmer,

⁵³ See *supra* Section B(1)(b) for further details on the different delegation models.

liquidity provider or crypto-lender wants to incur any liability for the other persons contributing the pool or for the pool itself.

The last argument generally defeats, for all intents and purposes, the characterisation of a crypto operation as a partnership. It is true that liability can be limited, such as in a Limited Liability Partnership (LLP), but this requires a registration of the entity. Since such formal registration is virtually non-existent for the operations analysed here, the partnership characterisation seems unpalatable for most cases. However, it cannot be ruled out that some courts might resort to this category when analyzing any of the operations covered here.

5. *Trust*

Trusts are well known within the realm of the Common law. In line with the purpose of this paper, we do not limit this legal analysis to specific national rules, but rather refer to a transnational notion. To this end, we refer to the notion of Trust retained by the Hague Trusts Convention,⁵⁴ as it aims to encompass any type of domestic forms of Trust-like arrangements, so long as some core characteristics are met. Like the drafters of the Convention, our aim is to focus on the skeleton and not to dissert on the changing colours of local variations of trusts. Moreover, and despite the relatively low number of States parties (14 in total to date),⁵⁵ the Hague Trusts Convention remains particularly relevant as some traditionally civil law countries (e.g. Italy, the Netherlands, Switzerland) have ratified it. Limiting ourselves to the very core and basic understanding of what a trust is in law (and especially in both English and American common law), we can restrict the legal qualification of trust to the following conditions: (1) There needs to be an outright transfer of assets, and (2) a settlor, a trustee, and beneficiaries must be identifiable.⁵⁶

⁵⁴ The Hague Convention of 1 July 1985 on the Law Applicable to Trusts and on their Recognition. Signatories include Australia, Canada, China, Cyprus, France, Italy, Liechtenstein, Luxembourg, Malta, Monaco, the Netherlands, Panama, San Marino, Switzerland, the UK and the U.S.

⁵⁵ The status table of the Hague Trust Convention can be found at <www.hcch.net/en/instruments/conventions/status-table/?cid=59> (last visited 22 December 2022).

⁵⁶ *Alfred E. von Overbeck*, Explanatory Report, International Legal Materials, Vol. 25, 1986, p. 593 et seq., para 39.

More precisely, pursuant to Article 2 of the Hague Trusts Convention, a trust refers to the legal relationships created when assets have been placed under the control of a trustee for the benefit of a beneficiary or for a specified purpose; and has the following characteristics:

- (a) the assets constitute a separate fund and are not a part of the trustee's own estate;
- (b) title to the trust assets stands in the name of the trustee or in the name of another person on behalf of the trustee;
- (c) the trustee has the power and the duty, in respect of which he is accountable, to manage, employ or dispose of the assets in accordance with the terms of the trust and the special duties imposed upon him by law.⁵⁷

Taking stock of the above, it should be underlined that the word "trust" needs not be used as such. Substance should prevail over form. Again, the characteristics outlined in the Hague Trusts Convention ambition to fit also trust-like arrangements that might exist outside the common law world.⁵⁸ This is not to say, however, that stricter criteria as regards the form or substance of trusts beyond what the Hague Trusts Convention provides may not be set under national law.⁵⁹

It is worth noting also that the Hague Trusts Convention applies only to trusts created voluntarily and evidenced in writing.⁶⁰ Such conditions call for some brief comments. First, the expression "trusts created voluntarily" is to be understood in contradistinction with trusts created by the operation of law or following a judicial decision.⁶¹ The intent to create a trust does not, however, necessarily need to be express, as it may be inferred from the relevant factual circumstances.⁶²

⁵⁷ The Hague Convention on Trusts is thus not limited to 'Anglo-American' trusts solely, but also aims to encompass all institutions that meet the criteria of Art. 2, see *Alfred E. von Overbeck*, Explanatory Report, International Legal Materials, Vol. 25, 1986, p. 593 et seq., para 25 and 46.

⁵⁸ See *Alfred E. von Overbeck*, Explanatory Report, International Legal Materials, Vol. 25, 1986, p. 593 et seq., para 26. See for instance the case of the "fiducie" under French law, as set out in Art 2011 to 2030 Civil Code.

⁵⁹ *Alfred E. von Overbeck*, Explanatory Report, International Legal Materials, Vol. 25, 1986, p. 593 et seq., para. 82.

⁶⁰ Hague Convention on Trusts; on the interpretation of international treaties see Richard Gardin, *Treaty Interpretation* (2nd edn, OUP 2017).

⁶¹ *Alfred E. von Overbeck*, Explanatory Report, International Legal Materials, Vol. 25, 1986, p. 593 et seq., para. 49.

⁶² *Alfred E. von Overbeck*, Explanatory Report, International Legal Materials, Vol. 25, 1986, p. 593 et seq., para. 52.

Second, the requirement of a written form aims to fulfil only evidentiary purposes.⁶³ A Trusts may well exist, even if it is formed orally or tacitly.⁶⁴ Moreover, the written form is not restricted to paper. It is commonly admitted today by the law of various legal systems that electronic documents fulfil the written form requirement insofar as they are stored durably. Under the arrangements scrutinized in this paper, the evidence of a trust may therefore be sought after in the general terms and conditions of platforms, and possibly even in the code and meta-data of the smart contracts used if these are humanly readable and understandable.

One particular obstacle to the qualification of the arrangements described above in Section A as trusts may stem, however, from the difficulties to distinguish between settlors, trustees, and beneficiaries. In the case of direct staking for instance, crypto-holders would tend to be settlors, trustees, and beneficiaries at the same time. Even if the roles may be mingled to some extent,⁶⁵ such an overlap of statuses would in all likelihood defeat the trust qualification. The same cannot be said of the staking-with-an-intermediary model, where the intermediary would play the role of a trustee. One difficulty arises, however, from the fact that crypto holders would qualify both as settlors and beneficiaries. Nevertheless, it is undisputed that crypto-assets are transferred outright by crypto holders to the platform intermediary, and that such intermediary would manage and stake the crypto-assets for the benefit of the former. The remaining question that only a case-by-case analysis could answer would then be whether the crypto-assets of the holders are comingled with the platform's own assets. If yes, the qualification of trust must fail. If, however, the crypto-assets received by the platform are held separately, the trust qualification might apply. In other words, where staking arrangements involve an intermediary as described in Section A above, the trust qualification cannot be discarded without further detailed analysis.

Yield farming may *prima facie* present itself as a hard case, since the platform intermediary would simultaneously be the trustee and the ultimate beneficiary of the arrangement. Indeed, it should be reminded that yield farming arrangements purport to stabilize the value of the crypto-assets issued and managed by the platform itself, and this for its own benefit. The broad transnational notion of trust enshrined in the

⁶³ Ibid.

⁶⁴ Ibid.

⁶⁵ *Alfred E. von Overbeck*, Explanatory Report, International Legal Materials, Vol. 25, 1986, p. 593 et seq., para. 47.

Hague Trusts Convention can nevertheless accommodate such a situation, as Article 2 paragraph 3 expressly sets that the fact that the trustee may himself have rights as a beneficiary is not necessarily inconsistent with the existence of a trust. Additionally, it is worth recalling that the transnational notion of trust under the Hague Trusts Convention entails that trustees may act for a specific purpose.⁶⁶ Thus, even if the yield farming platform ultimately fends for itself as the main beneficiary of the arrangement, it still arguably acts in accordance with and for the fulfilment of the specific purpose of the yield farming arrangement, which is principally the stabilization of its crypto-asset protocol. Moreover, it can be argued that a protocol with stable tokens also ultimately benefit their holders, who receive interest payments and have a right to reclaim assets whose value they helped to stabilize. Therefore, the very nature of yield farming arrangements does not irremediably defeat the trust qualification. Also, yield farming arrangements imply that crypto-assets are transferred outright. Yet, as for the case of staking-with-an-intermediary arrangements, the crucial question that a case-by-case analysis must answer is whether the crypto-assets received by the platform are comingled or not with its own assets.

A similar line of reasoning applies to liquidity mining, and similar conclusions may be drawn. In the case of crypto lending, however, a specificity should be underlined: As described in Section A above, crypto-assets of the initial holders are not generally segregated from the assets of the platform. As already mentioned, such characteristic would defeat the trust qualification, which leads to the conclusion that crypto lending arrangements are likely to fail to qualify as trusts.

The consequence of a trust characterization for risk bearing is the following: The customer (staker) as the settlor of the trust completely transfers any legal rights in the assets to the trustee who is, nevertheless, constrained by the trust to exercise legal powers in accordance with the trust. Critically, if the trustee goes bankrupt, the assets impressed with trust will not form part of his insolvency estate, as the beneficiaries will be generally able to require that they are removed from the insolvent trustee's estate and transferred to a new trustee. Thus, the particular attractions of a trust characterization in any insolvency: it tends to makes the assets at least seem 'insolvency-proof.'

⁶⁶ Art 2(1) Hague Trusts Convention.

6. *Investment Fund*

A layman may also envisage to characterize a pool of crypto-assets as a mutual fund. The problem with this regulatory term is that this commonly used designation masks a variety of different legal constructions under private law.

Within the United States, mutual funds are regulated by federal law in the Investment Company Act of 1940.⁶⁷ The Act focuses on the regulation and supervision of the fund but is generally of little relevance when it comes to the private rights of the investors. In particular, it is misleading to derive from this Act the notion “investment company” and use it as if it were a specific legal form.⁶⁸ Instead, the private rights of investors are governed by the law of the federal states. In most cases, funds are organized as trusts under state law. Managers have certain fiduciary duties towards investors, who enjoy certain rights in the assets of the fund.⁶⁹ Similar legal rules apply to UK open-ended unit trust, which are equally trusts, as the name indicates.

There is also a second model in which funds can be organized in the U.S., which can be called “the corporate model”. This model involves a corporation run by directors that has title to the assets. The investors, in turn, are the shareholders of the corporation. Despite the separate corporate personality, the directors have trustee-like duties that run directly to the investors.⁷⁰ A very similar model is followed by the Open End Investment Company (OEIC) in the UK, which accounts for about 50% of UK funds.⁷¹

All taken together, funds in two of the most important Common law countries are either trusts or subject to a trust-like regime, which makes them “trusts in disguise”.⁷²

In Europe, where mutual funds are known as “investment funds”, the legal situation is even more complex. The law of the European Union (EU) contains two

⁶⁷ 15 U.S.C. §§ 80a-1 to 64.

⁶⁸ Charles E. Jr. Rounds & Andreas Dehio, *Publicly-Traded Open End Mutual Funds in Common Law and Civil Law Jurisdictions: A Comparison of Legal Structures*, 3 N.Y.U. J.L. & Bus. 473-518, 478 (2006), who call the terminology of the Investment Company Act of 1940 “unfortunate”.

⁶⁹ *Id.* at 483.

⁷⁰ *Id.*

⁷¹ *Id.*

⁷² *Id.* at 478-479.

texts, the UCITS Directive⁷³ and the AIFM Directive⁷⁴. Similarly to the Investment Company Act of 1940, these directives concern the regulation and supervision of funds only. The private rights are subject to the national laws of the EU member states. We focus here on German law. Insofar, three models have to be distinguished: the contractual model, the co-proprietary model and the corporate model. The contractual model involves a fund that is owned by a managing company, with which the investors have a contractual relationship, but no rights in the assets. In the co-proprietary model, their situation is better in that they also have a share of ownership in the fund's assets. Finally, under the corporate model, the investors are shareholders of a company, that very similarly to the Common law corporate model owns the assets and has certain duties to the shareholders.

In sum, the terms “mutual funds” and “investment fund” do not help in the analysis of the private rights and obligations that arise from staking and similar operations. Instead, they mask a variety of the categories that have been previously analysed. It is therefore of no use to follow this category any further.

From a legal viewpoint, it is crucial whether the operation involves a loss of control over the crypto-assets or not.

7. Summary

All things considered, there are some legal categories that lend themselves better than others. The crypto market is, however, rapidly evolving, and the characteristics of the operations described may change accordingly. Also, legal doctrine varies significantly around the globe. Therefore, we refrain from specifying a definite legal category for each operation, but merely suggest various options which might be open for discussion. Rather, we focus on the distribution of risks that will result from the different characterizations.

If the characterization as a secured transaction that creates a limited right *in rem* – as opposed to an outright transfer – is adopted, our staker remains the owner

⁷³ Directive 2009/65/EC of the European Parliament and of the Council of 13 July 2009 on the coordination of laws, regulations and administrative provisions relating to undertakings for collective investment in transferable securities (UCITS) (recast), (2009) OJ L 302/32.

⁷⁴ Directive 2011/61/EU of the European Parliament and of the Council of 8 June 2011 on Alternative Investment Fund Managers (2011) OJ L 174, 1.

of the crypto-assets. In case of insolvency of the intermediary, he could thus ask for the return of his assets or their transfer to another intermediary.

In contrast, if a transaction is characterized as a loan or as an irregular deposit, then the staker transfers property to the intermediary. Should the latter go bankrupt, he is merely a creditor who must register his claim in the insolvency proceedings.

Where a transaction is seen as a (general) partnership, the assets would be jointly held between the intermediary and all 'staking' customers. At the same time, however, all parties involved would assume full liability for the debt of the enterprise.

Finally, in case of a trust characterization, the staker would no longer be legal owner of the assets that are now held by the trust. At the same time, he would not run the risk of insolvency of the trustee.

Drawing these strands together, two broad conclusions may be drawn. First, the protection of the staker depends on the transaction he was engaged in. If this transaction amounts to a transfer of assets, his risk in the event of insolvency of the intermediary is likely to be much higher. However, second, the precise risk cannot be predicted because it is subject to the legal characterization of the transaction by a court. The same operation may be viewed very differently, with the consequence of diverging risk profiles. For example, even if the *lex fori concursus* concludes that, for the purposes of characterisation, the issue in dispute is one of, say, loan contract; which then leads to the application of the conflicts rule for loan contracts; it will still be the law to which the conflicts rule points that ultimately determines whether the parties have concluded a valid contract of loan. In this sense, a conclusive determination of what the parties have achieved in law can only be made once the applicable law is known.

D. The Law Applicable to Crypto-Asset Transactions

As noted in Section A, the precise rules for identifying the applicable law differ from jurisdiction to jurisdiction. However, the traditional approach of 'conflict of laws,' sometimes known as 'private international law,'⁷⁵ in narrowing down one applicable law from many possibilities to govern a particular set of facts or legal relationship, is

⁷⁵ See Lawrence Collins & Jonathan Harris, *Dicey, Morris and Collins on the Conflict of Laws*, vol 1 (15th edn, Sweet & Maxwell 2012) para. 1-088 et seq.; Peter Machin North et al., *Cheshire, North & Fawcett Private International Law* 15-16 (OUP 2017).

to find some 'connecting factor' between the facts and a legal system. These include, for example: for property matters, the location of the property at the time of the disputed acquisition; for tort, the place where the damage was sustained; or in contract, the place of the characteristic performance of the contract. Whilst different connecting factors are used thus for different substantive areas of law,⁷⁶ ultimately, a common theme in the conflict of laws is the principle of the 'closest connection.'

A particularity of staking and similar transactions is that these connecting factors are often very difficult, if not impossible, to pinpoint. This is intentional and stems from the nature of DLT itself: The blockchain is designed to be a decentralized network; as such, it has no significant connection with any one specific state or place to the exclusion of all others. The same conclusion follows for any phenomena built on the blockchain, such as crypto-assets and smart contracts.

Accordingly, auxiliary criteria could instead be used to identify the legal jurisdiction with which DLT phenomena has the 'closest connection.' Such criteria might include the state in which most of the servers hosting nodes of the respective blockchain are located, or the location of the parties involved in a blockchain transaction. These auxiliary connecting factors, however, are equally difficult to locate themselves and, again, do not point to a single jurisdiction to the exclusion of all others. In addition, they might not have any particular relevance to the legal issue requiring determination.

These problems are compounded in the DeFi environment, where actors mostly are – and wish to remain – anonymous. Very few platforms are hosted or operated by an entity with legal personality. Even where the laws of a State permit the incorporation of a DAO, such option is rarely used. Instead, the typical construct in the DeFi environment is a suite of smart contracts stored on a blockchain. As such, it is impossible to find any connecting factor to link them to a particular legal system. Nevertheless, in the absence of a *lex cryptographica* or other autonomous legal system, legal disputes will be resolved by the application of a national law. Despite the tenuous nature of any connecting factors, however, a choice must be made.

⁷⁶ For an overview, see Michael Wilderspin, 'Contractual obligations' in Jürgen Basedow et al. (eds) *Encyclopedia of Private International Law* (Edward Elgar 2017) 473; Thomas Kadner Graziano, 'Torts' in Jürgen Basedow et al. (eds) *Encyclopedia of Private International Law* (Edward Elgar 2017) 1710; Louis d'Avout, 'Property and proprietary rights' in Jürgen Basedow et al. (eds) *Encyclopedia of Private International Law* (Edward Elgar 2017) 1429.

Accordingly, in this section, we consider the conflict-of-laws rules applicable to the transactions and legal relationships outlined in Section C above.

1. *The Applicable Law to Proprietary Issues*

When it comes to determining the law governing proprietary issues, two questions typically become important: The ownership of the crypto-asset in question and the existence of other, limited rights *in rem*.

a) Ownership of Crypto-Assets

Determining the law applicable to ownership of crypto-assets is arguably the most difficult present issue of conflict of laws.⁷⁷ Conflict of laws typically identifies as applicable the law of the state with whom a set of facts has the most significant connection. Finding this state creates headaches in a distributed environment where connections can be deemed to exist either to all countries or to none at all.⁷⁸ A way out may be to follow the principle of party autonomy, according to which the applicable law can be chosen. It is debated whether a choice of law in the context of property law is permissible.⁷⁹ If one answers this question in the affirmative, it remains to be settled how such a choice can be carried out. Yet these debates have little present practical value, since a choice of the applicable law in the crypto-context is – at least for the moment – very rare.⁸⁰

In the absence of a choice, connecting factors, such as the location of the private key, the blockchain, or the habitual residence of the programmer of the smart

⁷⁷ On this topic, see e.g. Andrew Dickinson, *Cryptocurrencies and the Conflict of Laws*, in David Fox and Sarah Green (eds) *Cryptocurrencies in Public and Private Law* (OUP 2019) para. 5.93-5.94; Michael Ng, 'Choice of Law for Property Issues Regarding Bitcoin under English law' (2019) 15 *Journal of Private International Law* 315-338.

⁷⁸ *Matthias Lehmann*, 'National blockchain laws as a threat to capital markets integration' (2021) 26 *Uniform Law Review* 168; Florence Guillaume, *Aspects of Private International Law Related to Blockchain Transactions*, in Daniel Kraus et al. (eds), *Blockchains, Smart Contracts, Decentralised Autonomous Organisations and the Law* (Edward Elgar 2019) 49.

⁷⁹ See Roel Westrik and Jeroen van der Weide (eds), *Party autonomy in International Property Law* (DeGruyter 2011).

⁸⁰ See however Art 3(2)(b) Liechtenstein TVTG; Sec. 12-107.(a)(c) UCC; Principle 5 UNIDROIT Digital Assets and Private Law Working Group, Study LXXXII - W.G.7 - Doc. 2, which allow for a choice of law in property law issues in the context of crypto-assets.

contract or blockchain protocol come into consideration.⁸¹ However, those are entirely unsuitable as they provide only a very tenuous connection, may point to several countries at a time and are subject to change. A more stable connection is provided by the law of the custodian. This factor is of particular importance given that most crypto-assets are held with an intermediary. The law chosen in the custody agreement, or, in the absence of a choice, the law at the place of business or incorporation of the custodian, can be used to determine the proprietary situation.

Apart from this, for those blockchains in which the providing of the network as well as the holding and transfer of crypto-assets is only possible after prior authorization, a choice of law for the network, e.g. in the protocol itself could be taken as a basis. In the absence of a choice, one could use the registered office or the habitual residence of the authority deciding on the admission to the blockchain as the factor determining the applicable law. However, this does not apply if an admission is required only the validation, but not for the participation in the network as such. In these cases, there is no sufficiently close connection between those persons who hold and transfer crypto-assets on the blockchain and that authority which decides on admission to the network. Instead, the law of the state of the registered office or habitual residence of the issuer of the crypto-asset in question is a suitable connecting factor. But what if none of these factors is present, e.g. in case of a directly held ETH that is staked to support the Ethereum network? For this case, other criteria will have to be used, connecting the network to a specific country. Inevitably, there will remain some situations that can hardly be connecting to any country, e.g. a bitcoin that is not held with a custodian that has been acquired in a peer-to-peer transaction.

In contrast, it is irrelevant for the determination of the law applicable to ownership whether the alleged owner of the crypto-asset is a consumer. While the classification as a consumer is relevant for the determination of the law applicable to contracts⁸², the conflict of laws rules for property matters are generally not linked to consumer status. Although the rules applicable to the contractual relationship can be consulted to determine the parties' intention as to whether a transfer of ownership is

⁸¹ See on those criteria Financial Market Law Committee (FMLC), Distributed Ledger Technology and governing law: Issues of legal uncertainty (2018), available at <http://fmlc.org/wp-content/uploads/2018/05/dlt_paper.pdf> (last visited 23 December 2022).

⁸² See *infra* Section D.2.

intended according to the will of the parties, they generally have no direct effect on the transfer of ownership as such.

b) Other Rights *in rem*

It is possible that the forum characterizes staking or a similar operation as a secured transaction.⁸³ This will lead to the creation of a right *in rem*. The conditions for the validity and the effects of such operation will depend on the conflicts rule of the forum for secured transactions.

Only very few jurisdictions have developed specific conflict-of-laws rules for secured transactions in crypto-assets. An example is Switzerland, which submits pledges to the law chosen by the parties or, in the absence of a choice, to the law of the state in which the pledgee has her habitual residence.⁸⁴ This rule, which traditionally applied to negotiable instruments, was recently extended to “similar rights”; this was with a view to treat digital token on an equal footing with their paper equivalents.⁸⁵

This rule is still an outlier. However, it is *a priori* not unreasonable and there are very few good alternatives to it. Theoretical and practical considerations suggest that other legislators might follow the same model.⁸⁶ This would mean that the validity and the effects of staking are determined in the first place by the law chosen by the parties. Such a choice is done, for instance, in the staking agreement.⁸⁷ In the absence of a choice, the law in force at the habitual residence of the staker governs.

If, as in most cases, the crypto-asset is held through an intermediary, or custodian, the additional question arises whether the law of the custodian has any incidence on the staking operation. Clearly, the law explicitly chosen to govern the staking agreement trumps any other law. But what if the staking agreement does not contain any choice of law? Can the staking operation then be submitted to the law governing the custody agreement or, in the absence of a choice in this agreement, the law at the place of business or the place of incorporation of the custodian? An

⁸³ See *supra* Section C.1.

⁸⁴ See Art 105(2) Swiss Federal Private International Law Act.

⁸⁵ See Swiss Federal Council, “Botschaft zum Bundesgesetz zur Anpassung des Bundesrechts an Entwicklungen der Technik verteilter elektronischer Register”, BBl. (Federal Gazette) 2020, p. 234, 297, available at: <<https://fedlex.data.admin.ch/eli/fga/2020/16>> (last visited 29 November 2022).

⁸⁶ In this sense Matthias Haentjens and Matthias Lehmann, *The Law Governing Secured Transactions in Digital Assets*, in Andrea Bonomi et al. (eds) *Blockchain and Private International Law* (forthcoming).

⁸⁷ See e.g. Coinbase User Agreement (January 2023), 9.5. Rewards, available at: <www.coinbase.com/de/legal/user_agreement/united_states> (last visited 23 December 2022).

argument can be made that a crypto-asset in custody becomes part of the legal environment of the custodian. This law is also easily identifiable for third parties. As a consequence, ETH in a portfolio held with a custodian that has been staked with a third staking service provider who has no choice-of-law clause in its terms would be governed by the law chosen in the custodian's terms and conditions.

2. *The Applicable Law to Contracts*

Contractual obligations are governed by the law that the parties to the contract have chosen. This follows from the principle of party autonomy, which is generally accepted around the world.⁸⁸

In the absence of a choice of law, different approaches are followed. In the US, many federal states apply a multi-factor test. It includes a variety of criteria, such as the domicile of either of the contracting parties, the place where the contract is concluded or the place of performance of the contractual obligations.⁸⁹

In Europe, the applicable law to contracts is that of the habitual residence of the party that must render the 'characteristic performance' of the contract⁹⁰. This will usually be performance of the non-monetary obligation. As a result, the law of the seller or service provider will be applied, instead of that of the buyer or customer.

In the following, it will be examined how these different connecting factors for determining the applicable law can be applied to the different types of transactions examined here.

a) Loan

In respect of loans, e.g. crypto lending,⁹¹ the applicable law can be normally chosen. An exception applies where the borrower is a consumer; in this case, the law at the place of her habitual residence applies. The same would be true for other borrowers as well where the applicable law has not been chosen. A court in Europe would apply the law of the lender because it is widely agreed that in the case of loans, it is not the borrower but the person granting the loan who provides the characteristic performance

⁸⁸ Symeon C. Symeonides, *Codifying Choice of Law Around the World* (OUP 2017) 113-115; Gilles Cuniberti, *Conflict of Laws: A Comparative Approach* (2nd edn, Edward Elgar 2022) 390.

⁸⁹ Restatement (Second) on the Conflict of Laws § 188(2) (1971).

⁹⁰ See for an overview Symeon C. Symeonides, *Codifying Choice of Law Around the World* (OUP 2017) 178-181.

⁹¹ See *supra* Section C.2.

of the contract.⁹² In the US, courts will have problems to apply the usual multifactor test to crypto-loans, given that the conclusion and the performance of the contract take place in cyberspace, with no discernible location.⁹³ Nevertheless, some US courts have reached the same result as those in Europe by favouring the law of the lender, in particular for larger transactions.⁹⁴

Since lending activities take place in chains, the applicable law can be different for each transaction. For instance, the law that applies to the contract by which crypto-assets are transferred to the pool of the crypto lending platform may differ from applying to the relationship between the crypto lending platform and the next borrower. This must be taken into account when analysing transactions that span multiple parties.

b) Deposit

Opposite results are achieved for the deposit. In this case, the characteristic performance is rendered by the person that *receives* the crypto, i.e. the custodian. Such a solution can be defended, e.g. for yield farming, and possibly also crypto lending, but rather not for staking.⁹⁵ The decision whether to characterize these operations as a secured transactions or a deposit is of double importance, both on the level of substantive law and on that of conflict of laws. Espousing a deposit characterization not only burdens the customer with the risk of insolvency of the intermediary, but also shifts the applicable law from the customer to the provider. In our view, this is an additional argument against this type of characterization.

c) Partnership

Where a crypto operation is characterized as a partnership, claims between the person providing crypto assets and the respective DeFi product on the one hand and claims between the individual persons providing crypto assets among each other on

⁹² CJEU, Case C-249/16, *Kareda v. Benkö*, 2017 ECLI:EU:C:2017:472, para. 41; Matthias Lehmann, 'Bonds and loans' in Jürgen Basedow et al. (eds) *Encyclopedia of Private International Law* (Edward Elgar 2017) 215, 217.

⁹³ See *supra* Section B.4.

⁹⁴ See e.g. *Gainer Bank, N.A. v. Jenkins*, 284 Ill.App.3d 500, 219 Ill.Dec. 809, 672 N.E.2d 317, 319 (1996); *Tuition Plan, Inc. v. Zicari*, 70 Misc.2d 918, 335 N.Y.S.2d 95 (1972); *Bowmer v. Dettelbach*, 109 Ohio App.3d 680, 672 N.E.2d 1081, 1082, 1085, 1086 (1996); *Pacific Gamble Robinson Co. v. Lapp*, 95 Wn.2d 341, 622 P.2d 850 (1980). However, a part of the literature assumes the location of the debtor to be decisive when determining the applicable law, cf. Eugene F. Scoles et al., *Conflict of Laws* (6th edn, West Group 2018) 1163.

⁹⁵ See *supra* Section C.3.

the other hand may be possible. Both types of claims are subject to the law governing the partnership as a whole.⁹⁶

There are two fundamentally different ways in how this law is to be determined. According to one approach, the law applicable to a partnership should be the law in which the partnership's office is located, i.e. in which the partnership has its effective place of management (seat theory). According to another view, only the place where the company was founded is relevant for determining the applicable law (incorporation theory).⁹⁷ These theories apply to all types of companies regardless of whether they are incorporated.⁹⁸ The determination of the applicable law to a non-registered partnership under both theories is subject to considerable difficulties.

On the incorporation theory, localization of the required incorporation act of the blockchain or DeFi platforms will not be possible as a rule. In general, DeFi platforms do not formally incorporate as a legal entity. The creation of the partnership takes place - depending on what exactly one considers this to be - as a rule on the Internet, with the launch of the blockchain or the acquisition of crypto-assets of the respective ecosystem. In all these cases, however, there is no formal registration of the respective act, which is why the place of incorporation is generally ruled out as a connecting factor. It is of course different for those DAOs that are legally incorporated in a certain jurisdiction.⁹⁹

If one instead resorts to seat theory to determine the applicable law, the specifics of the administration of blockchain or DeFi must be taken into account. Typically, these feature no centralized management; but decentralized governance by the crypto-asset holders voting directly on projects submitted to them. At first glance, the effective seat of administration could be determined by the place where the majority of the persons who hold crypto assets of the respective ecosystem have their habitual residence or registered office, or to the place where the majority of the crypto assets are held. However, for the vast majority of blockchains or DeFi platforms,

⁹⁶ for the U.S. see Eugene F. Scoles et al., *Conflict of Laws* (6th edn, West Group 2018) 1346.

⁹⁷ See on this from a European perspective Stephan Rammeloo, *Corporations in Private International Law* (OUP 2006) 95-96; from a U.S. perspective Eugene F. Scoles et al., *Conflict of Laws* (6th edn, West Group 2018) 1339-1344; Marc-Philippe Weller and Chris Thomale, 'Companies' in Jürgen Basedow et al. (eds) *Encyclopedia of Private International Law* (Edward Elgar 2017) 405-406.

⁹⁸ See Lawrence Collins and Jonathan Harris, *Dicey, Morris and Collins on the Conflict of Laws*, vol 2 (15th edn, Sweet & Maxwell 2012) para. 30-010, who do not distinguish between incorporations and partnerships to this extent.

⁹⁹ See e.g. for Vermont 11 V.S.A. § 4173 and for Wyoming Wyo.S.B. SF0038, 2021.

despite their decentralization, there is a group of persons, the so-called 'core team', approved by votes of the holders of the crypto assets who have programmed the algorithm and a front end in the form of a website, may suggest improvements, and provide support. These persons, who can be identified to varying degrees, thus take on a central function. They can thus be used to localise the blockchain or DeFi platform to a specific jurisdiction.

However, in many cases these persons are located in a large number of different countries or their place of residence is unknown. Identifying the applicable law hence may also run into factual difficulties. Therefore the decisive connecting factor should not be linked to individual persons or the majority of them, but to other factors, such as the place where coordination meetings are held or the location of premises that have been rented by the core team. However, very few platforms in the field of DeFi are likely to have such locations. Some blockchains or DeFi platforms also feature a person, typically called a curator, who has special powers. If the respective ecosystem provides for such a position, the effective seat of administration can also be located at the place of habitual residence of this person by means of this position. Yet also in this respect, there might be the problem that the spatial location of this person can be fraught with difficulties.

d) Trust

The conflict of laws faces the challenge of the trust being primarily a product of the Common law and therefore largely unknown to the legal systems following the Civil law tradition.¹⁰⁰ As a result, only a few Civil law jurisdictions have dedicated conflict-of-laws rules for trusts and instead, depending on the context in which the trust is used, fall back on the relevant conflict-of-laws rules, for example, on inheritance, family, debt or property law.¹⁰¹

Where legal systems provide for specific conflict-of-laws rules for trusts, these are primarily governed by a choice of law of the settlor.¹⁰² In the absence of such a

¹⁰⁰ Anatol Dutta, 'Trust' in Jürgen Basedow et al. (eds) *Encyclopedia of Private International Law* (Edward Elgar 2017) 1753.

¹⁰¹ Anatol Dutta, 'Trust' in Jürgen Basedow et al. (eds) *Encyclopedia of Private International Law* (Edward Elgar 2017) 1753, 1757.

¹⁰² Lawrence Collins and Jonathan Harris, *Dicey, Morris and Collins on the Conflict of Laws*, vol 2 (15th edn, Sweet & Maxwell 2012) para. 29R-001; See e.g. Art 6(1) Hague Trust Convention; Sec Restatement (Second) on the Conflict of Laws § 269(b)(i), § 270(a), § 271(a), § 272(a) (1971); Art 124 § 1(1) Belgian Private International Law Act.

choice, the place of administration of the trust designated by the settlor,¹⁰³ the location of the assets of the trust,¹⁰⁴ the domicile of the settlor,¹⁰⁵ the domicile or registered office of the trustee¹⁰⁶ and the places where the objectives of the trust are to be fulfilled¹⁰⁷ are applied.

If these principles are applied to staking via an intermediary, it becomes apparent that a large number of the connecting factors used in the dedicated conflict-of-laws rules cannot be determined without further ado. The most unambiguous is regularly the settlor's domicile and the trustee's domicile or registered office, even though those may be fraught with factual hurdles due to lack of knowledge of both's identity. For the place of administration of the trust, it may be possible to refer to the domicile of the intermediary conducting the staking. However, if the intermediary has no legal personality and there are no other indications as to its physical location, it is not possible to use this connecting factor. In turn, the registered office of the intermediary may be used to determine the location of the trust assets, if such an office exists. The crypto-assets themselves have no physical manifestation and ascription to a public key is recorded on a blockchain record, which in turn is decentralized accross servers around the world. The location of the private key, by means of which crypto-assets can be disposed of, is not a suitable connecting factor in this respect. Although the private key is necessary in order to dispose of the crypto-assets, it can be reproduced without further effort and thus cannot be equated with the location of the asset itself. A tying to the place where the objectives of the trust are to be fulfilled is also ruled out in the case of staking by means of an intermediary. The objective of the trust is the provision of crypto-assets for the validation of new blocks by means of a proof-of-stake mechanism. It may be possible to link this to the location of the server, which communicates with the blockchain network as a validator for this purpose. However, this would not do justice to the trust's objective of contributing to the validation of new blocks to the network as such. In this respect, the servers acting as validators are merely a means to achieve this objective.

¹⁰³ Art 7(2)(a) Hague Trust Convention.

¹⁰⁴ From a U.S. perspective Eugene F. Scoles et al., *Conflict of Laws* (6th edn, West Group 2018) 1275, 1278; Art 7(2)(b) Hague Trust Convention.

¹⁰⁵ See e.g. Restatement (Second) on the Conflict of Laws § 269(b)(ii), Sec 271(b) (1971); Art 124 § 2 Belgian Private International Law Act.

¹⁰⁶ Art 7(2)(c) Hague Trust Convention.

¹⁰⁷ Art 7(2)(d) Hague Trust Convention.

Overall, it thus emerges that the only connecting factors that work for a trust are the domicile of the settlor and the domicile or registered office of the trustee. Between these, the place of the trustee is generally preferred because of the overall role played by the trustee and his legal ownership of the assets. This would mean that the law applicable would be that at the domicile or registered of an intermediary with whom assets have been staked or to whom they have been transferred. In case the counterparty is a DeFi construct, however, this connecting factor fails as well and others have to be used. In this context, one may resort to the place of the 'core team' or similar considerations.¹⁰⁸

3. *The Applicable Law to Torts*

For the sake of completeness, we will also address the question of the law applicable to torts committed regarding staked cryptos, like 'rug pulls' or similar delicts.

It is possible for the parties to a claim in tort to choose the law applicable to the claim. However, such a choice is much more limited than a choice made in contract.¹⁰⁹ Typically, a distinction in torts is made between a choice made prior to the occurrence of the event giving rise to the claim, and a choice made after this event. The latter is largely uncontroversial and permissible without restriction. By contrast, a choice of law prior to the arising of the claim resulting from a tort is only possible under certain conditions, or is otherwise completely excluded.¹¹⁰ However, since the parties will often not be able to agree on a law to be chosen in the event of a dispute, the choice of law for torts is only of very minor importance.

In the absence of a choice of law, the determination of the law applicable to a tort is typically linked to a pre-existing relationship¹¹¹, to the common domicile of the

¹⁰⁸ See *supra* Section D.2.c.

¹⁰⁹ Kadner Graziano 'Torts' in Jürgen Basedow et al. (eds) *Encyclopedia of Private International Law* (Edward Elgar 2017) 1710, 1710 et seq.; Gilles Cuniberti, *Conflict of Laws: A Comparative Approach* (2nd edn, Edward Elgar 2022) 436; Symeon C. Symeonides, *Codifying Choice of Law Around the World* (OUP 2017) 40-41, 42, 51, 52-53.

¹¹⁰ Symeon C. Symeonides, *Codifying Choice of Law Around the World* (OUP 2017) 99-102.

¹¹¹ Thomas Kadner Graziano, 'Torts' in Jürgen Basedow et al. (eds) *Encyclopedia of Private International Law* (Edward Elgar 2017) 1710, 1712-1713; Eugene F. Scoles et al., *Conflict of Laws* (6th edn, West Group 2018) 806-807.

parties of the tort¹¹² or to the place of the tort, the *lex loci delicti*.¹¹³ This comprises both the place where the tort was committed and the place where the damage resulting from the tort occurred.

Due to the large number of tort claims potentially arising in connection with DeFi, it is not possible to address all potential situations on a case-by-case basis. Instead, the following is intended to describe in abstract terms the connecting factors potentially relevant in these cases as well as the difficulties in determining them. In doing so, it is necessary to consider, as a starting point, the typical factual circumstances in which tort claims may arise in connection with DeFi. Torts involving crypto-assets typically have at least three factual reference points which can be used to determine the applicable law: each of the two parties to the tort, and the smart contract into which the crypto-assets at issue are introduced. These points of reference provide a multitude of connecting factors, which in turn raise the question of localisation.

If the traditional connecting factors for non-contractual obligations arising in connection with DeFi described above are taken as a basis, identifying the applicable law is straightforward insofar as a legal relationship, like a staking agreement or an agreement on the deposit of crypto-assets, exists to which the determination of the applicable law can be linked. In these cases, the law applicable to this legal relationship is also decisive for the tort.

The place where the tort was committed may be help where it can actually be identified, e.g. where a hacker has intruded into a DeFi construct from his home. In other cases, it will however be impossible to determine. The question of determining the place where the damage occurred is trickier. For this place, it is already questionable from a legal point of view how precisely this place is to be determined. For instance, one could refer to the location of the crypto-assets, the location of one of the parties, the location of an intermediary or the location of the smart contracts by means of which the DeFi construct is realized. From a factual point of view, these

¹¹² Thomas Kadner Graziano, 'Torts' in Jürgen Basedow et al. (eds) *Encyclopedia of Private International Law* (Edward Elgar 2017); see Eugene F. Scoles et al., *Conflict of Laws* (6th edn, West Group 2018) 794-800, 801-804.

¹¹³ Thomas Kadner Graziano, 'Torts' in Jürgen Basedow et al. (eds) *Encyclopedia of Private International Law* (Edward Elgar 2017) 1710-1711; Gilles Cuniberti, *Conflict of Laws: A Comparative Approach* (2nd edn, Edward Elgar 2022) 436; Symeon C. Symeonides, *Codifying Choice of Law Around the World* (OUP 2017) 40-41, 42, 51, 52-53.

places often cannot be located. For example, what criteria should be used to determine the location of crypto-assets recorded on a decentralized blockchain? How can an intermediary be geographically located if it has no legal personality and no physical presence? Insofar as the place where the damage occurred depends on the localization of the DeFi construct, it has already been shown that this is possible in some cases under certain conditions, but as a rule one is confronted with almost insurmountable hurdles.

Realistically, a court will often not be in a position to determine the applicable law to a tort in the DeFi context using the connecting factors mentioned above. Instead, it could apply the law of the forum. However, this would incentivize claimants to shop for the forum that has the law most favourable to their claim. A preferable approach would be to follow the law at the place of domicile of the victim. This will result in the application of one and the same law regardless of where a claim is brought. Such an approach can be justified under the traditional approach prevailing in Europe if one is prepared to accept the victim's domicile as the 'place where the damage occurs'. There are some cases in which such a view has been accepted by the European Court of Justice, specifically with regard to torts committed online.¹¹⁴ This could provide the basis for a larger acceptance in other cases as well.¹¹⁵ The advantage of this view is that it protects the victim of rug pulls and other scams.

The determination of the applicable law for torts is therefore fraught with considerable difficulties in legal and factual terms, insofar as these cannot be linked to an existing contractual relationship. In view of the limited admissibility of a choice of law, such a choice also offers an adequate possibility for a legally secure determination of the applicable law only in the rarest of cases. Unless the law cannot be determined otherwise, the law of the victim's domicile should be applied.

¹¹⁴ CJEU, Case C-375/13, *Kolassa v. Barclays Bank*, 2015 ECLI:EU:C:2015:37, para. 55; CJEU, Case 509/09, *eDate Advertising and Others v. X and Société MGN LIMITED*, 2011 E.C.R. I-10269, ECLI:EU:C:2011:685, para. 49, 52.

¹¹⁵ See *Matthias Lehmann and Emeric Prévost*, Table ronde sur la methode de la localisation dans l'espace digital; Contribution II: «La localisation du dommage dans le Web3», (2022) *Revue de droit des affaires internationales / International Business Law Journal* 736.

E. Conclusion

This contribution has sought to explore the insolvency risks customers run when using cryptoassets to generate passive income through an intermediary. As has been demonstrated, there is considerable legal uncertainty on the rights a customer has on the intermediary's insolvency, which, absent legislation or authoritative case law, cannot be mitigated. Some general points can however be made.

First, it is first important to recognise that, whilst the term 'staking' is loosely understood in commercial practice to mean using cryptoassets as an investment asset, it is not a term of commercial art. This paper has demonstrated that 'staking' encompasses a wide range of practices that, furthermore, must be distinguished. This is primarily because, from a legal perspective, they give rise to a very different set of legally relevant facts – such as an outright transfer of the asset – and to different legal inferences as to the parties' intentions in respect of those facts.

Second, the rights of the customer on the intermediary's insolvency will depend on how a court will characterize these transactions. Characterization, in the first instance, is relevant for the conflicts rule that will ultimately be selected by the forum to determine the law governing the transaction. This governing law will, then, have the final say on the customer's rights upon the intermediary's insolvency: whether it retains some right in the asset such that it is removed from the insolvency estate entirely, whether ranks in preference to other creditors, or whether it must join the queue with the intermediary's other unsecured creditors for a *pari passu* share in whatever remains of the insolvency estate.

Finally, it is important to note that the conflicts rules are different from court to court. The applicable law thus ultimately depends to some degree on where the intermediary's insolvency is opened, i.e., the location of the bankruptcy court. As this cannot always be predicted, an element of uncertainty remains.

Nevertheless, it is possible to draw a spectrum of risks. It can be safely predicted, for instance, that a direct staking agreement is likely to involve the least risks for the customer in the event of insolvency of the intermediary, whereas crypto lending arrangement carries high risks for him. All other transactions are situated between these two extremes. Which risks precisely are incurred can be said only after the national law applicable to the operation has been identified.